

FMP · MODULE 7 · WORKBOOK

OCCUPATIONAL CERTIFICATE · FINANCIAL MARKETS PRACTITIONER

Foreign Exchange Markets workbook answers.

Model answers to the Module 7 practical-evidence workbook — all four questions, **90 marks**. Answers are drawn from Novia's own study guide and class deck, the workbook's case study, and the SARB *Currency and Exchange Manual for Authorised Dealers*.

Question 1 · The FX market, exchange rates and money	30 marks
Question 2 · Quotations, reciprocals and market players	25 marks
Question 3 · Millkorn CC — import payment & exchange control	10 marks
Question 4 · Case study — the 2013 forex fixing scandal	25 marks
Total	90 marks

1.1 Define the foreign exchange market and explain its significance in the global financial system. (3)

The foreign exchange market — also known as forex, FX or the currencies market — is an **over-the-counter (OTC) global marketplace that determines the exchange rate for currencies around the world**. Participants buy, sell, exchange and speculate on the relative exchange rates of various currency pairs, and currencies are **always traded in pairs**, so the value of one currency in the pair is relative to the value of the other. The market is made up of **banks, forex dealers, commercial companies, central banks, investment management firms, hedge funds, retail forex dealers and investors**.

Its significance is fourfold. First, it is **by far the largest and most liquid financial market in the world** — the study guide puts turnover at **over US\$5.2 trillion a day** — so it is the price-setting mechanism for every currency on earth. Second, it is a **global network of financial centres that transacts 24 hours a day, closing only on weekends**: as one hub closes another opens, which increases liquidity and adds to its appeal as the largest asset class available to investors. Third, it is the mechanism through which **international trade and foreign direct investment are settled** — every cross-border payment for goods, services or investment passes through it. Fourth, it allows participants to **hedge commercial and asset-portfolio currency risk**, to invest in currency as an asset class in its own right, and to arbitrage and speculate.

1.2 Differentiate between “free-floating” and “fixed float” currencies and provide examples of each. (3)

The value of a country's currency depends on whether it is a free float or a fixed float.

- **Free-floating currencies** are those whose **relative value is determined by free-market forces**, such as supply-and-demand relationships, with no government intervention setting the level.
Examples: the US dollar, the Japanese yen and the British pound.
- **Fixed float** is where a country's **governing body sets its currency's relative value to other currencies**, often by **pegging it to some standard**, and it is not allowed to fluctuate freely.
Examples: the Panamanian balboa and the Saudi riyal.

For completeness, the study guide identifies a **third regime** that sits between the two: a **dirty float or managed exchange rate**, where the rate is **influenced by government via the central bank around a preferred rate** rather than either fully free or rigidly fixed.

QUESTION 1 · THE FOREIGN EXCHANGE MARKET, CONTINUED

1.3 List and describe the three main types of foreign exchange markets. (3)

MARKET	DESCRIPTION
Spot forex market	The immediate exchange of currencies at the current exchange rate — “on the spot” . It makes up a large portion of the total forex market and involves buyers and sellers from across the entire spectrum of the financial sector, as well as individuals exchanging currencies. In practice foreign exchange settles two business days after the dealing date.
Forward forex market	An agreement between the buyer and the seller to exchange currencies at an agreed-upon price at a set date in the future. No exchange of actual currencies takes place, just the value. The forward market is often used for hedging .
Futures forex market	Like the forward market in that there is an agreed price at an agreed date . The primary difference is that the futures market is regulated and happens on an exchange , which removes the risk found in the other markets . Futures are also used for hedging.

1.4 Identify and briefly explain four factors that influence currency exchange rates. (3)

The foreign exchange market is very volatile, and the study guide identifies a number of influences. Four of the most important are:

- **Inflation** — inflation **devalues a currency because it decreases the currency's buying power**. A country with persistently higher inflation than its trading partners will tend to see its currency weaken.
- **Interest rates** — rates set by central banks **affect the cost of borrowing and the yield on holding a currency. Higher interest rates attract foreign capital and can lead to currency appreciation; lower rates lead to depreciation.**
- **Political stability** — **stable political conditions encourage foreign investment, which leads to appreciation**, and a stable currency is perceived as a safe store of value. **Political instability, corruption or economic uncertainty deters investors.**
- **Imports and exports (the trade balance)** — when a country's **exports exceed its imports it creates a trade surplus**, which strengthens the economy and increases the value of its currency. A country importing more than it exports can see inflation and a weaker currency.

Other influences named in the guide include **government debt** and the country's debt rating, **market sentiment, economic growth**, and **natural disasters, wars or technological advancements**.

1.5 What is Purchasing Power Parity (PPP), and how is it used in economic analysis?

Purchasing power parity (PPP) is a popular **macroeconomic analysis metric used to compare economic productivity and standards of living between countries**. It is an economic theory that compares different countries' currencies through a “**basket of goods**” approach: PPP is **the exchange rate at which one nation's currency would have to be converted into another in order to purchase the same amount of the same large group of products** — in other words, the rate at which the same products and services can be bought in each country. (3)

It is used in economic analysis in three ways. The **PPP exchange rate is used to convert the local currency of a target nation into a common currency applicable to all nations** — normally the US dollar, or what is called the **international dollar**, a currency designed to serve as a baseline. **Some countries adjust their gross domestic product (GDP) figures to reflect PPP**, so that economies can be compared on like-for-like purchasing power rather than at market exchange rates. And it allows analysts to compare **standards of living** across borders. The caveat is that **some feel PPP does not reflect reality, because of differences in local costs, taxes, tariffs and competition**.

Worked example from the study guide

A shirt costs **\$10** in the U.S. and **€8.00** in Germany. To make an apples-to-apples comparison, the €8.00 must first be converted into U.S. dollars. If the exchange rate is such that the German shirt costs **\$15.00**, then:

$$\text{PPP} = 15 \div 10 = 1.5$$

In other words, for every **\$1.00** spent on the shirt in the U.S., it takes **\$1.50** to obtain the same shirt in Germany when buying it with the euro.

1.6 Explain the concept of exchange rates and how the foreign exchange market originated. (3)

The concept. An exchange rate is the price of one currency expressed in terms of another. The foreign exchange market is an **over-the-counter global market where the buying and selling of global currencies occurs, determining their exchange rates**. It is made up of banks, forex dealers, commercial companies, central banks, investment management firms, hedge funds, retail forex dealers and investors, **all of whom trade currency pairs**. Because currencies are always traded in pairs, an exchange rate is always **relative** — it states how much of the **quote currency** is needed to purchase one unit of the **base currency**.

How it all began. As long as humans have been trading there has been a foreign exchange market: **ancient civilisations traded goods and currencies through metal coins whose value was based on their weight**. The **first true forex market was in Amsterdam, approximately 500 years ago**, where the exchange allowed people to freely trade currencies to stabilise exchange rates. In **1875 the gold standard was implemented**, meaning countries could only print currency equal to the amount of their gold reserves — gold being rare, malleable, tough to corrode and hard to obtain. By **1913 there were 71 forex trading firms in London, up from three in 1903**. The gold standard **could not hold up during the world wars**, as countries had to print money to finance expenses. **After World War II the Bretton Woods system was established**, calling for most currencies to be pegged to the U.S. dollar, which was itself backed by gold reserves. In **1971 President Nixon announced a freeze on the dollar's convertibility to gold**, and in **1973 the gold standard was completely abolished** — the dollar was no longer backed by gold and foreign exchange switched to a **free-floating system**, with currencies free to peg or remain unpegged. Finally, **the Maastricht Treaty was signed in 1992** to introduce a single European currency, and **the euro was launched on 1 January 1999**.

1.7 Differentiate between fiat currency and representative money.**(3)**

The main difference is **what stands behind the money**: **fiat currency is backed by the government's trust, while representative money is backed by a commodity.**

	FIAT CURRENCY	REPRESENTATIVE MONEY
Backing	The government's trust . It is not backed by a commodity with intrinsic value ; the government's official recognition of it as a medium of exchange is what gives it value .	A commodity, such as gold or silver , and it can be exchanged for that commodity .
Form	Coins, banknotes or digital currencies — physical money, paper or coins.	A cheque or other form of currency that can be exchanged for physical money in a stated amount ; it can also represent the intent to pay , such as a cheque or debit card.
Example	The US dollar became fiat money in 1971 , when the country left the gold standard.	The gold standard is the notable example — each country's currency tied to a fixed amount of gold.

Both fiat and representative money possess the value they claim to have. The distinction is only in what guarantees that value — a government on the one hand, a commodity on the other.

1.8 Discuss three factors that influence the supply and demand of a currency. (3)

Supply and demand for any given currency — and thus its value — is **not influenced by any single element but by several**, which generally fall into three categories: **economic factors, political conditions and market psychology**. Three of the factors the study guide identifies are:

- **International trade** — **when a country exports more than it imports, the demand for its currency increases**, because foreign buyers must acquire that currency to settle for the goods. The reverse applies when a country is a net importer, since residents must sell local currency to buy foreign currency.
- **Inflation** — inflation is **the rate at which a currency loses its purchasing power**. As purchasing power falls, the currency becomes less attractive to hold, demand for it falls, and its value declines relative to lower-inflation currencies.
- **Interest rates** — **in a growing economy the demand for money increases, which raises interest rates**. Higher rates in turn attract foreign capital seeking that yield, which raises demand for the currency and supports its value.

The guide also names **economic growth** (a country's economic prosperity determines its population's purchasing power), **domestic economic downturns** (when a currency is not in demand its value falls relative to other currencies), and **capital flow and money supply**. It notes that **the most common method to value currency is through exchange rates**.

QUESTION 1 · CONTINUED

1.9 Explain the difference between currency appreciation and depreciation and their impact on imported goods. (3)

	CURRENCY APPRECIATION	CURRENCY DEPRECIATION
Definition	A currency's value increases relative to another currency.	A currency's value decreases relative to another currency.
Effect on prices	Lower effective price for imported goods.	Higher import prices.

Put in rand terms: if the rand **appreciates** — say USD/ZAR moves from 7.11 to 6.98 — a South African importer needs **fewer rand to buy each US dollar**, so the landed cost of imported goods falls and **inflation decreases**. If the rand **depreciates** — USD/ZAR moves from 7.11 to 7.20 — the importer needs **more rand per dollar**, so **import prices increase and inflation increases**.

Importantly, **there is no optimal value for a currency**. A stronger currency does not necessarily benefit everyone: a **strong rand makes a South African importing company more competitive but a South African exporting company less competitive**, and the reverse holds for a weak rand. A stronger currency **benefits some people and sectors in the economy but hurts others — it is not necessarily better, it is just different**.

1.10 What are ISO currency codes, and how are they used in forex trading? (3)

ISO currency codes are three-letter alphabetic codes that represent the various currencies used throughout the world, and each country-specific three-letter code **also has a corresponding three-digit numeric code**. The codes are identified by the **International Organisation for Standardisation (ISO)**, a non-governmental organisation that provides standards for manufacturing, commerce, technology and communication; for currencies **the governing document is ISO 4217:2015**.

In forex trading they are used to build currency pairs: **when combined in pairs they make up the symbols and cross rates used in currency trading**, and they are **central to currency pairs, which are the quotation and pricing structures of the currencies traded**. **The first three-letter code in a quotation is the base currency and the second is the quote currency**, and the pair indicates **how much of the quote currency is needed to purchase one unit of the base currency**.

Example and the major pairs

EUR/USD is the quote for the euro against the U.S. dollar. A quoted price of **1.2500** means that **one euro is exchanged for 1.25 U.S. dollars**, because EUR is the base currency and USD is the quote (or counter) currency — another way of looking at it is that **it will cost you \$125 (USD) to buy €100 (EUR)**.

The major pairs are **EUR/USD** (euro / U.S. dollar), **GBP/USD** (Great Britain pound sterling / U.S. dollar), **USD/JPY** (U.S. dollar / Japanese yen), **USD/CHF** (U.S. dollar / Swiss franc) and **USD/ZAR** (U.S. dollar / South African rand) — implying that **the USD is the base currency and the ZAR is the quoted or second currency**.

2.1 Explain the concept of “bid and offer” in the spot foreign exchange market using the USD/ZAR quotation as an example. Highlight the difference for a bank and a client. Assume the current exchange rate is 7.25/7.26. (5)

As in other markets where banks quote a two-way price, **banks also make a two-way price in foreign exchange**. When a bank — the **price maker** — quotes a price, it indicates to its counterparty — the **price taker or price user** — the rate at which it is prepared to **buy (bid)** and the rate at which it is prepared to **sell (offer)** the **base currency** against the **variable or quoted currency**. The price user then “**selects**” whether to buy or sell the base currency at the quoted price. The critical convention is that **exchange rates are always quoted from the bank’s point of view**.

In the quotation **USD/ZAR 7.25 / 7.26**, the USD is the base currency and the ZAR is the variable currency. The **bid is 7.25** and the **offer is 7.26**.

USD/ZAR 7.25 / 7.26	BID — 7.25	OFFER — 7.26
The bank price maker	Buys US dollars at 7.25 Sells rand at 7.25	Sells US dollars at 7.26 Buys rand at 7.26
The client price user / counterparty	Sells US dollars at 7.25 Buys rand at 7.25	Buys US dollars at 7.26 Sells rand at 7.26

The difference for the bank and for the client

The two sides are **mirror images**: whatever the bank does, the client does the opposite at the same rate. The bank **buys the base currency low** — at the “smaller price” or “cheaper rate” of **7.25** — and **sells the base currency high** — at the “bigger price” or “more expensive price” of **7.26**. The client is therefore always on the worse side of the quote: a client selling dollars receives only **R7.25** per dollar, while a client buying dollars must pay **R7.26** per dollar.

In practical South African terms: a **South African importer** needing dollars to pay an overseas supplier **buys the currency from the bank at the offer, 7.26**, and the bank sells it. A **South African exporter** receiving dollars from an overseas customer **sells the currency to the bank at the bid, 7.25**, and the bank buys it.

THE SPREAD — A POINT THE GUIDE UNDERLINES The difference between the bid and offer rates is known as the **spread**. Here it is $7.26 - 7.25 = 0.01$, **one cent per US dollar**. The study guide is explicit that this spread **does not represent a dealer profit but a “risk spread”** — it compensates the market maker for the risk of holding a position, since it is unlikely the market will deal on both prices simultaneously and dealing on one side leaves the trader “**taking a position**”.

QUESTION 2 · CONTINUED

2.2 Differentiate between direct and indirect currency quotations with examples. Assume the following rates Rand 7.5050 and Sterling 1.7575. (5)

In the foreign exchange market a **variable number of units of one currency is quoted per one unit of another currency**. Which currency sits in the base position is what separates a direct quotation from an indirect one.

	DIRECT QUOTATION	INDIRECT QUOTATION
Base currency	The US dollar	The local or domestic currency
What it expresses	The amount of local currency needed to buy or sell 1 US dollar	The amount of US dollars needed to buy or sell one unit of the local currency
Example given	1 USD/ZAR = 7.5050 — one US dollar equals R7.5050	GBP/USD = 1.7575 — one British pound equals 1.7575 US dollars
Reading it	You need R7.5050 to obtain US\$1 . The rand is the variable (quoted) currency.	You receive US\$1.7575 for £1 . Sterling is the base and the dollar is the variable (quoted) currency.

Market convention. Most currencies in the foreign exchange market are **quoted against the US dollar as the base currency** with the local or other currency as the quoted currency — these are the **direct quoted currencies**. However, certain currencies are **conventionally quoted as the base against the US dollar as the variable currency**: the **euro**, **sterling (GBP, £)**, the **Australian dollar (AUD, \$)** and the **New Zealand dollar (NZD, \$)**. When quoting against the euro, it is market practice in the interbank market to quote all currencies in terms of a varying number of units **per 1 euro** — the euro is **by market convention always the base currency**.

Applied to South Africa: **all exchange rates are quoted against the US dollar, and all other currencies are cross rates**. So R7.5050 is a direct quotation for the rand, while £1 = US\$1.7575 is an indirect quotation for sterling. If the same principle were applied to sterling as to the rand, the GBP/USD quotation implies that **one British pound equals 1.7575 US dollars**.

2.3 Calculate the reciprocal bid and offer rates for a given USD/ZAR spot rate of 6.8525/6.8575. Show your calculations. (5)

The principle. Any quotation of a currency pair with a particular currency as the base can be converted into the equivalent quotation with that currency as the variable or quoted currency — that is, **a direct quoted pair can be converted into an indirectly quoted pair and vice versa**. This is done by **taking the reciprocal ($1 \div \text{the rate}$) of each side**.

Calculation

USD/ZAR bid $\rightarrow 1 \div 6.8525$ **= 0.145932...**

USD/ZAR offer $\rightarrow 1 \div 6.8575$ **= 0.145826...**

Resulting ZAR/USD quotation **0.1458 / 0.1460**

The study guide works this identical example and publishes the answer as **ZAR/USD 0.1458 / 0.1460**.

THE POINT THE MARKS ARE FOR — THE SIDES REVERSE The smaller number is always on the left and the higher number on the right, so the two sides of the quotation are reversed. The reciprocal of the USD/ZAR **bid** (6.8525) becomes the ZAR/USD **offer**, and the reciprocal of the USD/ZAR **offer** (6.8575) becomes the ZAR/USD **bid**. In either case the convention is unchanged: **the price maker will buy the base currency on the left against the variable currency and sell the quoted currency on the right**.

Note on rounding. Carried to six decimals the two reciprocals are **0.145826** and **0.145932**, which round to **0.1458 / 0.1459**. The study guide publishes **0.1458 / 0.1460**, rounding the upper side out to the wider figure. I have shown the exact arithmetic and quoted the guide's published answer, since that is the answer the guide teaches.

QUESTION 2 · CONTINUED

2.4 What are the key advantages of using spot exchange rates for businesses? List and explain any three. (5)

ADVANTAGE	EXPLANATION
Simplicity	The spot exchange rate is straightforward enough to understand — it is simply the price at which one currency can be exchanged for another currency on the spot date . There is no interest-rate differential, forward point or optionality for the business to interpret.
Flexibility	It is a flexible option for businesses that need to exchange currencies short term , because it allows companies to take advantage of favourable exchange rates as they arise rather than being locked into a rate agreed months earlier. It also lets participants transact without a long-term commitment .
Speed	Transactions are completed quickly, typically within two business days . This is a real advantage for companies needing to pay for goods or services promptly — for example an importer settling a cash-with-order supplier.
Cost-effectiveness	The spot rate is generally a cost-effective option for businesses needing to exchange small amounts of currency, as it does not require any upfront payment or deposit — unlike a futures contract, which requires margin.

The study guide also lists **real-time pricing** — the rate **reflects current market conditions and is updated in real time**, allowing companies to make informed decisions about their currency exchanges — and a **transparent environment**, in that **spot markets facilitate trading where transactions occur at prevailing prices that are public information**.

2.5 Discuss the different players in the foreign exchange market and their roles. (5)

The FX market is a global marketplace where currencies are exchanged against each other. Participants use it for a variety of reasons, including **hedging against currency and interest rate risk, diversifying portfolios, and speculating on geopolitical events**. The study guide identifies the following players:

PLAYER	ROLE
Central banks	Considered the market's stabilizers — they intervene to influence the level of the currency and to implement monetary policy.
Commercial banks	Considered the core of the market . They make the two-way prices and execute international payments on behalf of customers.
Hedge funds	Known for their strategy expertise .
Sovereign wealth funds	Considered the silent giants — very large but infrequent participants.
Forex dealers	Provide access to the market for other participants.
Corporations	The practical users of the market — importers, exporters and multinationals converting currency for real commercial need.
Proprietary trading firms	Known for taking risks , trading their own capital.
Brokers	Facilitate transactions in the market between counterparties.
Liquidity providers	Considered the backbone of the market .
Speculators	Known for moving prices .
Individual traders	A small portion of the market , but often use it for speculation and day trading .

THE SOUTH AFRICAN OVERLAY Most banks in South Africa have been registered as **Authorised Dealers (ADs)**, trading in forex, by the Treasury department of the **SARB**, and a list of ADs appears in **Section A.2(A) of the Currency and Exchange Manual for Authorised Dealers**. The **FSCA** is the market regulator that oversees forex trading in South Africa. The local stock exchanges — **JSE Limited, A2X Markets, ZAR X, CTSE and EESE** — trade only in **currency derivatives such as currency futures**, which are **standardised exchange-traded contracts**.

The facts

Millkorn CC, a South African company, imports machinery to supply the local and SADC sawmilling industry. Its new supplier in **Guangzhou, China**, deals **only on a cash-with-order / advance basis in US dollars**. The order is for **kilns for drying and heat treating at a cost of USD 94,000.00**, for a local customer in Mpumalanga.

3.1 Which payment method will Millkorn CC have to consider?

(1)

Cash in Advance — payment in advance. The supplier deals only on a cash-with-order basis, so Millkorn must pay the full USD 94,000 **before shipment of the goods**. Of the four main international payment methods — cash in advance, open account, documentary collections and documentary credit (L/C) — this is the only one the supplier will accept.

3.2 List at least TWO (2) risks that Millkorn CC is facing, when paying for goods in advance.

(2)

Payment in advance **carries the lowest risk for the seller but the highest risk for the buyer**.

Millkorn's risks include:

- **Non-delivery of the goods** — the **goods may never be shipped**. Millkorn has parted with USD 94,000 and holds no documents of title and no security whatsoever.
- **Incorrect or defective goods** — the guide's own wording is that **incorrect goods may be received**. Having already paid in full, Millkorn has **no commercial leverage** to force a correction and would have to pursue the supplier in a foreign jurisdiction.
- **Foreign exchange / currency risk** — the rand cost of the USD 94,000 is fixed only once the spot deal is done; between quoting its Mpumalanga customer and paying the supplier, **a depreciating rand raises the landed cost** and erodes Millkorn's margin.
- **Loss of the use of working capital** — the cash is out of the business for the whole manufacturing and shipping lead time, worsening its operating cycle, while the supplier is paid before it even begins production.
- **Supplier insolvency or fraud**, and **country / political risk** in the supplier's jurisdiction.

3.3 Payment will be made against a “pro-forma invoice”, how will Millkorn CC clear the goods at point of arrival (i.e. customs)?

(2)

A **pro-forma invoice is not a commercial invoice** — it is a quotation in invoice form used to arrange the advance payment, and **it is not acceptable to customs for clearing purposes**. To clear the goods on arrival Millkorn must obtain from the supplier the **final commercial invoice** together with the **transport document (bill of lading or air waybill)**, the **packing list** and any **certificate of origin**, and lodge a **customs declaration (the SAD 500) with SARS Customs** — normally through its clearing and forwarding agent — supported by those documents, paying any duty and VAT due.

The same documents serve a second purpose under exchange control: because the payment was made **in advance, before the goods existed**, the **Authorised Dealer must obtain documentary proof that the goods were in fact imported**. Millkorn must therefore provide its bank with the **customs declaration / proof of importation** within the period required by the Currency and Exchange Manual, so the AD can acquit the advance payment.

QUESTION 3 · CONTINUED

3.4 Section B.1(D) of the Currency and Exchange Manual for AD covers the Terms of Payment for imports. Explain the difference between Capital Goods and Consumer Goods. Provide at least ONE (1) example of each. (4)

	CAPITAL GOODS	CONSUMER GOODS
What they are	Goods that are used to produce other goods or services — plant, machinery and equipment that form part of the productive capacity of a business.	Goods bought by the end user for personal or household use and consumed rather than used to produce something else .
Use	Held and used over a long useful life ; not resold in the form in which they are bought.	Consumed or used up , usually within a short period.
Lead time	Typically manufactured to order with long production and shipping lead times , which is why the terms of payment permitted for them differ.	Generally off-the-shelf, standard and quickly available .
Example	The kilns for drying and heat treating that Millkorn is importing — also sawmill machinery, a factory plant, or commercial vehicles.	Food, clothing, and household appliances such as a television or a kettle.

The distinction matters under **Section B.1(D), Terms of Payment**, because the manual applies **different advance-payment conditions to each**: because capital goods carry long manufacturing lead times, Authorised Dealers may permit payment to be made **further ahead of shipment** than for consumer goods, subject in each case to the AD **obtaining documentary evidence that the goods were actually imported**.

3.5 Considering the goods that Millkorn CC is importing, under which section of the Exchange Control Ruling will it be paid? Motivate your answer. (1)

Section B.1 — Imports, and specifically **B.1(D), Terms of Payment**, dealing with **advance payments for capital goods**. The motivation is that Millkorn is a **South African resident importing physical goods**, paying a **foreign supplier in foreign currency through an Authorised Dealer**, and the kilns are **capital goods** — plant used to produce other goods — being **paid for in advance of shipment**, which is exactly the transaction B.1(D) governs.

WORTH CHECKING AGAINST THE LIVE MANUAL Section and sub-section lettering in the SARB *Currency and Exchange Manual for Authorised Dealers* is amended from time to time. The substance above — imports under Section B.1, terms of payment under B.1(D), and the AD's duty to obtain proof of importation — should be confirmed against the current version downloaded from the SARB website, as the workbook instructs.

4.1 How did the 2013 forex scandal come to light?

(2)

It came to light through the press. **Bloomberg News reported in June 2013** that currency dealers said they **had been front-running client orders and rigging the foreign exchange benchmark WM/Reuters rates** by colluding with counterparts and pushing through trades before and during the 60-second windows when the benchmark rates are set. Following that report, **market regulators in Asia, Switzerland, the United Kingdom and the United States began to investigate** the \$5.3 trillion-per-day market.

The case study adds that the behaviour **occurred daily in the spot foreign-exchange market and went on for at least a decade**, and that although **suspicious movements in the prices of major currencies indicated that something was wrong**, those suspicious price movements were — embarrassingly for the managers responsible — **first highlighted by a whistleblower** rather than by internal controls.

4.2 How were the exchange rates rigged?

(2)

The rigging exploited the design of the daily fix. Because the fix was **based on actual transactions over a very short period of time** — originally a window of **30 seconds before and 30 seconds after 16:00 London time** — **the potential existed for market players to get together and place orders during that 60-second window**. If those orders were **big enough they could affect the benchmark calculation and create profit opportunities for their firms**.

Mechanically, **senior currency traders at five of the biggest banks colluded through online/electronic chatrooms** — with exotic names such as **The Bandits Club, The Cartel and The Mafia** — to place **aggressive “buy” or “sell” orders, known in the business as “banging the close”, in order to distort the fix**. In those chatrooms traders **disclosed confidential customer order information and trading positions, changed trading positions to accommodate the interests of the collective group, and agreed on trading strategies** as part of a group effort to manipulate the benchmark rates. The chatrooms were **exclusive and invitation only**, and membership was highly sought after.

4.3 Was there a regulatory failing? Motivate your answer.

(2)

Yes — there was a clear regulatory failing. The motivation runs as follows.

- **The market had been left largely unregulated** because, ironically, **regulators considered the forex market “too big to be manipulated”**. That assumption proved wrong.
- **There were early warning signs that all was not well. Minutes of a meeting of dealers at the Bank of England in 2006 appear to suggest that the possibility of market manipulation was discussed in front of officials — although the Bank of England denies this interpretation.**
- **Regulators have themselves accepted the point: they now know that light-touch regulation was an invitation to the financial services industry to game the rules, and they have responded with more intrusive supervision and hefty deterrents.**
- **Nine years on from that 2006 meeting, the Bank of England has led global regulators in cleaning up the forex market — “and not before time, critics will say”.**

That said, the failing was **not the regulators' alone**. The case study is equally clear that **clues available to outsiders should have been picked up internally long ago**, that **managers meant to have been in charge of the traders** failed to detect it, and that **prime responsibility lies with those who participated directly**.

4.4 Can such scandals be prevented?

(2)

They can be suppressed, at least for a period, but not permanently guaranteed against. The case study's assessment is that **institutional cheating of the kind seen in the Libor and forex scandals will probably die out for a while**, for four reasons: **individual traders have seen colleagues marched off the trading floor to face questioning; managers have finally understood the need for line-by-line, desk-by-desk scrutiny; regulators have moved from light-touch regulation to more intrusive supervision and hefty deterrents; and structural reforms have been made — the fix window has been extended from one minute to five, and the Bank for International Settlements has sought a unified code of conduct.**

Against that background **it would be surprising if systemic malpractice were to continue in the immediate future**. But the conclusion is a warning rather than an assurance: **there is no room for complacency in an industry where corporate memories are short and the rewards for beating the market are great.**

4.5 By how many minutes has the daily 4pm fix been extended to? (1)

The window in which the daily 4pm fix is calculated has been **extended from one minute to five minutes** — an increase of **four minutes**, on the recommendation of the **Financial Stability Board**. **This makes it harder to manipulate**, because a colluding group must now move a far larger volume over a far longer window to shift the benchmark.

4.6 What was the estimated amount that was lost due to the manipulation of the forex market? (1)

As of **December 2014**, the monetary losses caused by manipulation of the forex market were estimated to represent **\$11.5 billion per year (£7.5 billion per year) for Britain's 20.7 million pension holders alone**. The case study stresses that **the manipulations affected customers all around the world, for over a decade**, and that **the overall estimated cost is not yet fully known**.

4.7 Which FIVE banks were fined by the Financial Conduct Authority (FCA)? (5)

On **12 November 2014** the United Kingdom's **Financial Conduct Authority imposed fines totalling \$1.7 billion on five banks** for failing to control business practices in their **G10 spot foreign exchange trading operations**:

BANK	FCA FINE
Citibank	\$358 million
HSBC	\$343 million
JPMorgan	\$352 million
RBS — Royal Bank of Scotland	\$344 million
UBS	\$371 million
Total	\$1.7 billion

On the **same day** the United States **Commodity Futures Trading Commission (CFTC)**, in coordination with the FCA, imposed **collective fines of \$1.4 billion against the same five banks** — \$310 million each for Citibank and JPMorgan, \$290 million each for RBS and UBS, and \$275 million for HSBC. On **20 May 2015** the five banks **pleaded guilty to felony charges** brought by the US Department of Justice and agreed to pay fines **totalling more than \$5.7 billion**; Barclays, not part of the November 2014 settlement, was fined **\$2.4 billion**, UBS also pleaded guilty to wire fraud with a \$203 million fine, and Bank of America — **while not found guilty** — agreed to a \$204 million fine for unsafe practices. Barclays was fined a further **\$150 million on 18 November 2015** for automated electronic foreign exchange misconduct.

4.8 What was the finding about managing risk?

(2)

The FCA determined that **between 1 January 2008 and 15 October 2013 the five banks failed to manage risks around client confidentiality, conflict of interest, and trading conduct**, and more broadly **failed to control business practices in their G10 spot foreign exchange trading operations**.

The consequence of that failure was that **the banks used confidential customer order information to collude with other banks to manipulate the G10 foreign exchange currency rates and profit illegally at the expense of their customers and the market**. The remediation that followed went to the same root cause: the FCA's programme **required firms to review their IT systems in relation to their spot FX business, as the banks relied on legacy technologies that allow for the existence of dark-data silos within which manipulation is able to occur unnoticed by compliance systems**.

4.9 What is your understanding of the G10 currencies? List them.**(5)**

My understanding is that the **G10 currencies are the ten most heavily traded and most liquid currencies in the world**, issued by large developed economies with **stable governments, credible central banks and freely convertible, floating currencies**. Together they account for the overwhelming majority of global foreign exchange turnover, which is precisely why the scandal is described as affecting “**G10 spot foreign exchange trading operations**” — the manipulation struck at the core of the market, not at a peripheral corner of it. The name is historic and comes from the group of countries that agreed to participate in the IMF's General Arrangements to Borrow; despite the label, it refers to **ten currencies**.

CODE	CURRENCY	ISSUER
USD	US dollar	United States
EUR	Euro	Eurozone
JPY	Japanese yen	Japan
GBP	Pound sterling	United Kingdom
CHF	Swiss franc	Switzerland
CAD	Canadian dollar	Canada
AUD	Australian dollar	Australia
NZD	New Zealand dollar	New Zealand
SEK	Swedish krona	Sweden
NOK	Norwegian krone	Norway

Note that **four of these — EUR, GBP, AUD and NZD — are quoted by market convention as the base currency against the US dollar**, while the rest are quoted with the USD as the base. The South African rand is **not** a G10 currency; in South Africa all rates are quoted against the US dollar and everything else is a cross rate.

4.10 Is the Forex Market regulated in South Africa? Motivate your answer. (2)

Yes. Although the global forex market has been **largely unregulated** — regulators having considered it “**too big to be manipulated**” — South Africa regulates foreign exchange at **two distinct levels**, and the motivation rests on both.

- **Market conduct regulation.** The **Financial Sector Conduct Authority (FSCA)** is the market regulator in South Africa that oversees forex trading. It is responsible for ensuring that financial service providers in South Africa are regulated and adhere to established financial sector legislation. The study guide's point is that trading with FSCA-regulated forex brokers ensures South African traders enjoy a high level of security.
- **Exchange control.** South African residents are bound by South African exchange control laws and are not permitted to speculate against the rand. In terms of the Exchange Control Laws, **only Authorised Dealers (ADs) and Authorised Dealers with Limited Authority (ADLAs), appointed by the South African Reserve Bank, are permitted to buy, sell, borrow or lend foreign currency.** Every cross-border conversion must therefore pass through a regulated intermediary applying the *Currency and Exchange Manual for Authorised Dealers*.

So South Africa is materially **more tightly regulated than the global OTC market described in the case study**: conduct is policed by the FSCA, and the flow of currency itself is policed by the SARB through the exchange control framework.

4.11 Who authorises dealing in foreign exchange transactions in South Africa? (1)

The **South African Reserve Bank (SARB)**. Most banks in South Africa have been registered as **Authorised Dealers (AD)**, trading in forex, by the Treasury department of the SARB, and it is the SARB that appoints both **Authorised Dealers** and **Authorised Dealers with Limited Authority (ADLAs)**. The list of Authorised Dealers is published in **Section A.2(A) of the Currency and Exchange Manual for Authorised Dealers**.